C1.4 How can scientists help improve the supply of potable water?	
Teaching and learning narrative	**Assessable learning outcomes** *Learners will be required to:*
The increase in global population means there is a greater need for potable water. Obtaining potable water depends on the availability of waste, ground or salt water and treatment methods. Chlorine is used to kill microorganisms in water. The benefits of adding chlorine to water to stop the spread of waterborne diseases outweigh risks of toxicity. In some countries the chlorination of water is subject to public regulation, but other parts of the world are still without chlorinated water and this leads to a higher risk of disease (IaS4).	1. describe the principal methods for increasing the availability of potable water, in terms of the separation techniques used, including the ease of treating waste, ground and salt water including filtration and membrane filtration; aeration, use of bacteria; chlorination and distillation (for salt water) 2. describe a test to identify chlorine (using blue litmus paper) *PAG2*

Linked learning opportunities

Ideas about Science:
- technologies to increase the availability of potable water can make a positive difference to people's lives (IaS4)
- access to treated water raises issues about risk, cost and benefit and providing treated water for all raises ethical issues (IaS4)

Practical work:
- identify unknown gases